

```
C:\>Python27\Scripts\pip install ibmiotf
```

The command prompt window is as shown in Fig. 5.

One important function used in Python code (bluemix_iot.py) is def myfn(). It is used for collecting data from the IP address using HTTP. Tempvalue is used as a variable to hold data. The given code publishes the last value of captured data on the cloud at all pre-determined intervals.

Python sensor gateway is the frontend software installed on the PC having an Internet connection via Wi-Fi. As shown in Fig. 2, Mbed LPC1768 MCU is connected to the PC/laptop via its LAN port. Python program (bluemix_iot.py) has all the functions to read sensor data from the sensor through the Ethernet port of Mbed NXP LPC1768 board.

Please note that Internet connection on the PC through Wi-Fi is preferred without any proxy restrictions. During testing, the Internet connection was obtained through a Wi-Fi hotspot on a smartphone.

After the hardware is set up, upload the firmware to Mbed LPC1768 board and install Python sensor gateway. The next step is to configure IBM cloud platform and stream sensor data on the cloud.

Configuring IBM Bluemix cloud platform

The steps for configuring IBM Bluemix Cloud platform are given below.

Go to www.bluemix.net and click on Create an IBM Cloud account to create a free account. Fill in your valid email ID, name, country, phone number and password. You will find confirmation for the account in your email. Open your email and click on the link sent during registration. (Please remember that the free account with this email will be valid for one month only.)

Login to IBM cloud and select Catalog. A window will appear, as shown in Fig. 6.

Click Starter Kits→Internet of Things Platform Starter.

Enter App name (for example, mytestiot123) and click on Create from the Windows, as shown in Fig. 7.

The dashboard will look like as shown in Fig. 8. Select sk1234 name.

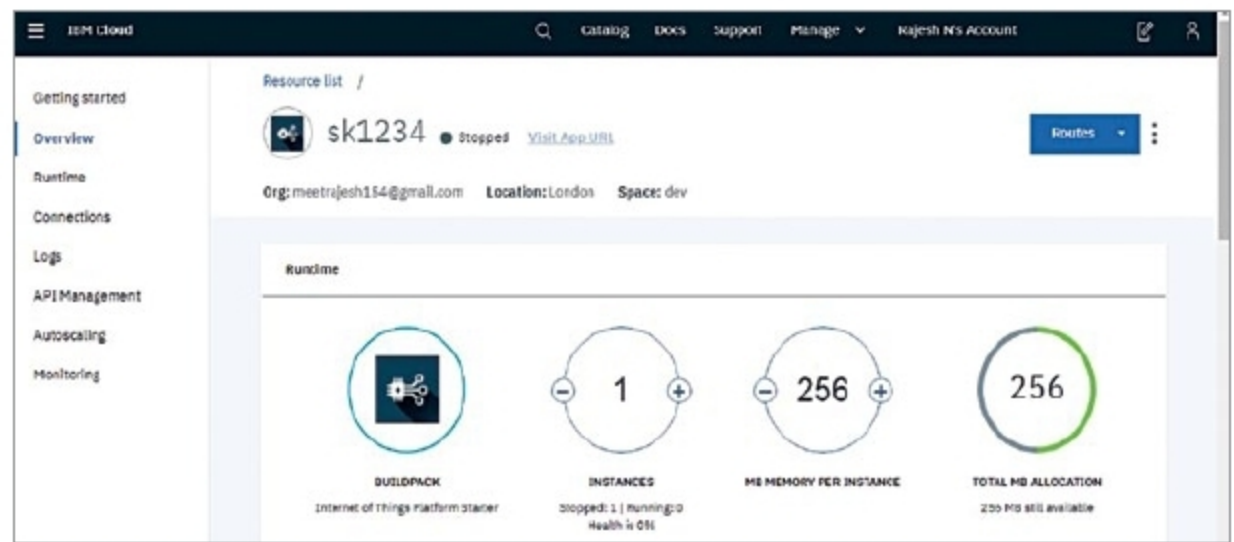


Fig. 8: Dashboard

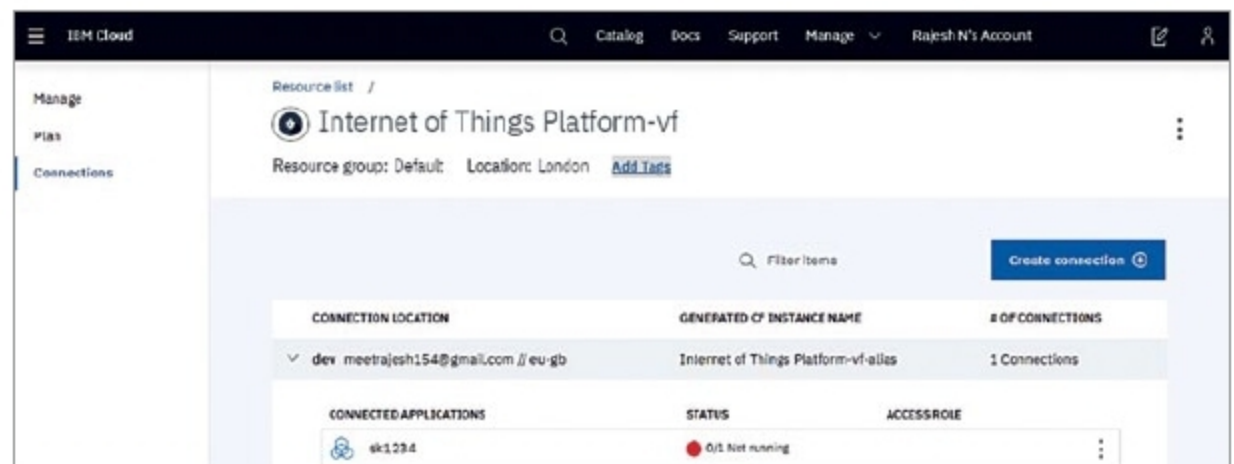


Fig. 9: Internet of Things Platform

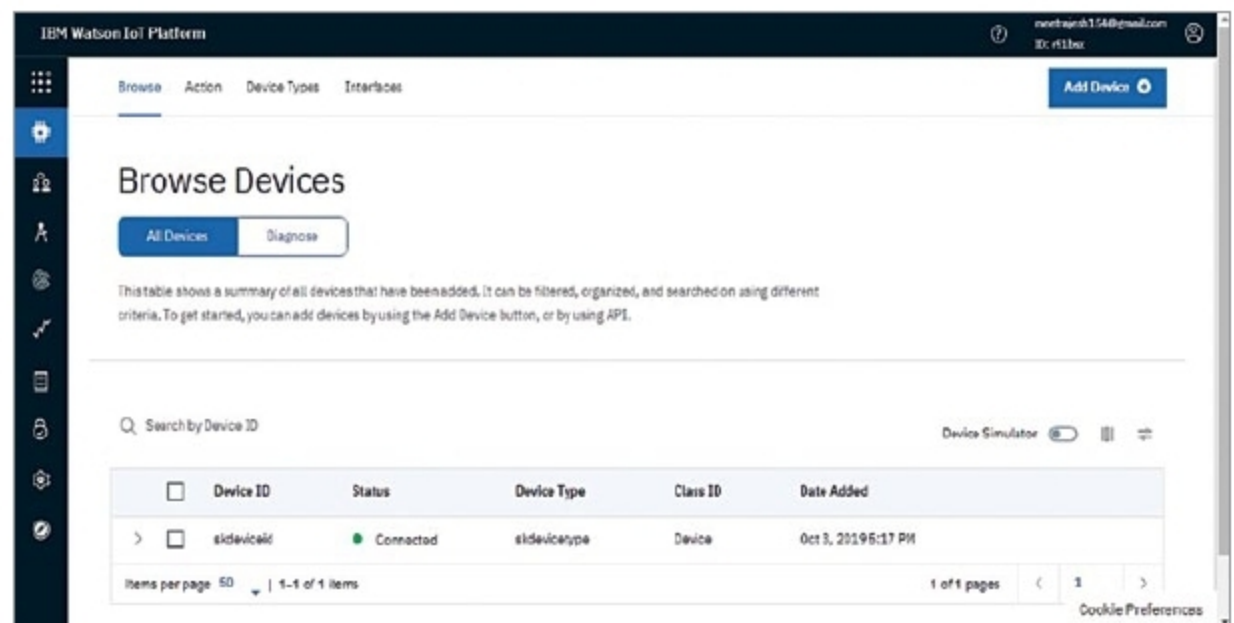


Fig. 10: Add Device button

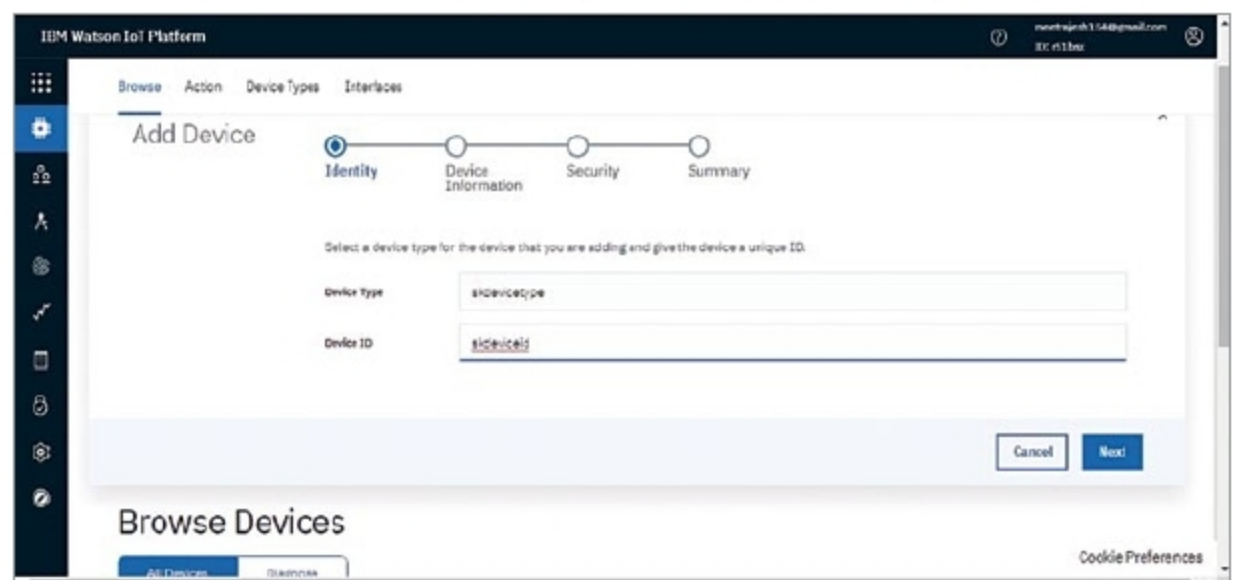


Fig. 11: Device ID and type